

Sponsored Search and User Attention: A Conceptual Exploration of Influence and Trust

Jashan Jot Kaur Gill¹, Manav Gupta², Tarun Bhardwaj³, Rajat Takkar⁴

¹²³⁴*Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, 140401*

¹jashanjot1559@gmail.com, ²manavgupta2678@gmail.com, ³tarunbhardwaj016@gmail.com,
⁴rajat.takkar@chitkara.edu.in

Abstract: Search engines retrieve relevant information in the form of hyperlinks, text, images and videos based on the user query. Modern search engines transform a search engine result page (SERP) into a mix of organic links (merit-based results) and sponsored links (paid advertisements). The interaction differences between organic and sponsored links are determined by the nature of the user's query. Sponsored links are the primary source of revenue for the search engines. However, within the sponsored links, the click-based optimisation only considers the behavioural response of the user, while the attitudinal response is largely overlooked. The user experience is shaped by the relevance, trust, content and visual placement of the results. This paper conceptually examines how sponsored search affects user attention (Danesco-Niculescu-Mizil et al., 2010), trust formation (Petty and Cacioppo, 1986), and decision-making behaviour (Lu et al., 2017), evaluating both the limitations and benefits of sponsored results. This paper aims to explore the consequences of visibility bias and examine how cross-platform advertising may enhance the organic rankings of a website, which may lead users to perceive the ranking as purely merit-based. This highlights the need for increased transparency, relevance-sensitive ranking and an ethical interface design that keeps user autonomy and trust in contemporary search ecosystems.

Keywords: Search engine result page · Sponsored link · Organic link · Search engine · Search engine optimisation · User experience · Click through rate · Expected revenue per mille impressions · Platform ecosystems · Cross-platform advertising · Branded search behaviour

1 Introduction

The development of web search has led to greater access to information. With the widespread use of the World Wide Web, web searches have become increasingly monetised. Search engines are the most commonly used portals to locate e-commerce websites. Approximately 89% of consumers use search engines for purchase decisions (Griwet, 2012), and 71% of business purchase decisions start with SEs (Meloni, 2011). The popularity of SE attracts e-commerce companies. SERPs are primarily composed of two types of information, organic results(merit-based) and sponsored results(ads). E-commerce websites must enhance their search engine visibility to remain valid in a highly competitive business environment. Moreover, SEs need to earn revenue from the content they provide users for free. The SE advertising provides a middle ground through which SE can generate revenue, and advertisers can promote their products. In SE advertising, advertisers bid on terms they presume their customers would type in the search box. Along with each bid term, advertisers specify the matching type (Exact, Phrase, or Broad) they want associated with the term (Aiello et al., 2016).

Although SE advertising benefits both SEs and advertisers, it is doubtful that it benefits the SE users as well. A survey published by [Nielsen.com](https://www.nielsen.com) (Nielsen 2015) to assess the extent to which people trust a certain form of advertising found that people consider ads from traditional media more credible than ads from SEs, website banners and mobile phones. Further studies also reveal the users' bias against Sponsored Links (SLs). Users need to believe that SLs provided by the SE do have the competence and integrity to provide unbiased, useful and credible information. A study conducted to examine the bias against SLs by switching OLs and SLs found a significant preference for OLs (Jansen et al., 2007). Searchers viewed OLs first more than 82% of the time. A positive correlation was reported between brand familiarity and the reputation of the ad and consumers' trust (Chaudhuri and Holbrook, 2001; Grabner-Kraeuter, 2002; Szczepanski, 2005).

There are some fundamental differences between the Organic Links (OLs) and SLs. For every user query, OLs aim to satisfy the most likely user intent, and SLs focus on the potential commercial intent (Danesco-Niculescu-Mizil et al., 2010) as inferred from the user query. While OLs are ranked based on

relevance, SL ranking considers both revenue and relevance. This paper conceptually explores how cross-platform advertising through social media platforms can reshape user behaviour and influence the organic rankings of an e-commerce website.

The rest of this paper is organised as follows. Section 2 discusses the theoretical background and related literature. It introduces the platform ecosystem perspective and the impact of cross-platform exposure on search visibility. Section 3 describes a conceptual framework combining attention allocation, trust formation, relevance, and ecosystem dynamics to explain users' interaction with SLs. Section 4 examines the benefits, structural tensions, and ethical issues related to sponsored search systems and encourages transparency and user experience. Section 5 includes implications for search engine operators, advertisers, and policymakers. Section 6 talks about the study's limitations and Section 7 wraps up the paper by summarising key insights and suggesting a more transparent and relevance-focused vision for sponsored search.

2 Theoretical Background and Related Literature

2.1 The Architecture of Sponsored Search

Sponsored search aims to show an advertisement that does both gives users what they were looking for and profits the SE. In SE advertising, the advertisers bid on related terms they presume users might search for; these SLs are then displayed on either the top (north ads) or the left (east ads) of the OLs (Danescu-Niculescu-Mizil et al., 2010). Ranking among the SLs is typically done by revenue optimisation, measured using expected revenue per mille impressions (eRPM), which is calculated by the product of click through rate (CTR) and the bid amount of the advertiser (Aiello et al., 2016). However, such methods create a bias towards the advertisers with rich click histories, and highly relevant advertisements with insufficient click history might never get a chance.

Along with targeted keywords or bidden terms, advertisers have to provide the matching type (exact, broad, phrase) for each of the keywords. While OLs' "snippets" are algorithmically generated, advertisers craft ads manually. Advertisers have to provide for the advertisement a title, body, text, and display URL. Both the OLs and the SLs provide users with information, but with different objectives (Danescu-Niculescu-Mizil et al., 2010). While SLs focus on the potential commercial intent and are ranked by combining relevance and revenue, OLs aim to provide the users with the most relevant content.

A large proportion of query-advertisement pairs receive very few impressions in ranking models based entirely on eRPM (Aiello et al., 2016). This results in disproportionate click data history among many advertisers. This may alter the perception of how well an advertisement addresses the user's information needs. Advertisements with rich click history are thus preferred over the newer, potentially more relevant ones. This creates incumbency effects that can distort the diversity and quality of sponsored results.

2.2 Relevance as a Distinct Dimension from Clickability

Clickability in itself is not enough to represent relevance (Aiello et al., 2016). It was noted that many advertisements with a high CTR are not necessarily relevant, and those with a low CTR might be highly relevant to the user query. There was a remarkable amount of advertisements (60%) that were labelled by the editors as "highly relevant" that did not receive any clicks (i.e., $CTR = 0$) (Aiello et al., 2016). This implies that if the ad serving model relies on eRPM to rank advertisements, 60% of "highly relevant" ads may still not get a chance to be displayed due to their low CTR. Similarly, 17% of advertisements with relatively high CTR are judged as "Irrelevant". This means the ranking system negatively impacts the user experience by excluding content users might find useful and surfacing content that fails to address the user's needs.

A supervised machine learning model based entirely on textual features (Cosine similarity, BM25 score, Jaccard coefficients, LSI representation, and hash embedding features) is proposed to address the limitations, independent of the click history (Aiello et al., 2016). Throughout the dataset used, a Random Forest classifier trained on 185 such features outperformed both the basic 19-feature text baseline ($AUC = 0.667$) and a click augmented baseline ($AUC = 0.595$) by achieving an AUC of 0.764. Since the model is independent of click data, it can be applied to new advertisements with insufficient click history and also enable several practical applications, such as pre-filtering irrelevant SLs, improving click model accuracy, recovering high-relevance SLs lacking prior data, and relevance boosted re-ranking.

Online A/B tests showed that relevance-based filtering improved CTR for exact match SLs by 4.07% while reducing revenue by only 1.71%. Incorporating relevance into the click model improved the click yield by 3.02%, and recovering high-relevance SLs without click data produced a 3.70% revenue uplift.

These findings suggested that explicit relevance modelling is not merely a compromise between user experience and commercial objectives but supports both simultaneously.

2.3 Organic–Sponsored Interplay: Competition and Synergy

Using a dataset of 63,789 query-organic-sponsored result tuples collected over a month from Yahoo! search, it was concluded that there exists a non-monotonic relationship between OLs' and SLs' CTR (Danescu-Niculescu-Mizil et al., 2010). For navigational queries, a clear competitive relationship was observed: high organic CTR was associated with low sponsored CTR, as the top organic result typically satisfied the user's need, leaving little incentive to engage with sponsored alternatives. For non-navigational queries, the relationship became positively correlated: higher organic CTR was associated with higher sponsored CTR, suggesting that when queries were informational or transactional, both result types collectively address the user's more open-ended information need.

The study also demonstrated a persistent bias against sponsored results: even when sponsored ("North") ads and organic results led to the same domain—a near-equal-quality setting—users preferred organic results in 52% of cases, with sponsored CTR averaging only 95% of organic CTR— notable given that North ads were displayed above the organic results, suggesting position alone does not override users' preference for non-commercial content. Consistent with prior studies (Lu et al., 2017), this pattern is partly attributed to user scepticism towards commercially motivated content, highlighting the role of trust in shaping user interaction with sponsored results.

A further contribution of the study (Danescu-Niculescu-Mizil et al., 2010) is the distinction between responsive and incidental advertisements. Responsive ads directly address the user's information need and closely match the query intent reflected in the organic results. In contrast, incidental ads are only tangentially related, reflecting what advertisers expect users interested in the query topic might also find useful. Their findings reveal that for navigational queries, incidental ads outperformed responsive ones in terms of CTR. Because navigational queries are typically fully satisfied by the top organic result, a responsive ad offers little additional value and struggles to overcome the existing bias against SLs. However, incidental ads introduce complementary information not present in the OLs, providing users with a reason to engage. The opposite pattern emerges for non-navigational queries. Incidental ads are perceived as less beneficial, as they diverge too far from the user's broader, open-ended information needs.

2.4 Dual-Process Trust Theory Applied to Sponsored Search

The dominant theoretical framework for understanding user trust in sponsored links is the dual-process model, derived from social psychology (Lu et al., 2017). This framework, when applied in the context of search engine advertising, distinguishes between two routes through which trust attitudes are formed in response to SLs: the central route and the peripheral route.

The central route is activated when users possess both the motivation and the ability to process relevant information (Lu et al., 2017). Users form their trust through deliberate cognitive assessment (Liu and Shrum, 2009; Tam and Ho, 2005). The formation of cognitive trust depends on whether the trustee has the necessary attributes to rely upon (Komiak and Benbasat, 2004). Through this route, users form cognitive trust, defined as a rational assessment of the competence and integrity of advertising models in providing useful and unbiased results. The factors affecting the cognitive trust are the perceived contextual relevance of the SL to the user query, the reputation of the SE hosting the SL, and the user's understanding of how the SL operates. Contextual relevance of sponsored links was found to be the strongest predictor of cognitive trust ($\beta = 0.713$, $p < 0.01$), followed by search engine reputation ($\beta = 0.189$, $p < 0.01$) in a $2 \times 2 \times 2$ factorial laboratory experiment involving 178 participants (Lu et al., 2017).

The peripheral route to trust is used to process information when the user lacks either the motivation or the ability to engage in cognitive processing. Considering the lack of ability of an e-commerce user to interact with the seller and experience the product, individuals tend to form trusting attitudes via peripheral routes. Search engine reputation significantly predicts emotional trust ($\beta = 0.342$, $p < 0.01$), as does the inverse of ad avoidance tendency ($\beta = 0.345$, $p < 0.01$), indicating that users with habitual aversion to online advertising tend to exhibit lower emotional trust regardless of content quality.

Both cognitive and emotional trust predict attitudinal and behavioural responses. Attitudinal responses are measured by the accuracy of recall of SL content, while behavioural responses are measured as total clicks. Cognitive trust exerts stronger effects on both outcomes (attitudinal: $\beta = 0.361$; behavioural: $\beta = 0.339$) compared to emotional trust (attitudinal: $\beta = 0.317$; behavioural: $\beta = 0.303$). Notably, knowledge of the sponsored link mechanism does not significantly increase cognitive trust

($\beta = -0.005$, $p = 0.163$), suggesting that disclosure of the advertising mechanism alone is insufficient to build trust when relevance quality is not simultaneously addressed.

A further finding demonstrated through a YES-NO recall test suggests that SLs, even when not clicked, generate measurable attitudinal effects, including brand recall and attitude formation (Lu et al., 2017). Participants were asked to recall SL content regardless of whether they had interacted with those links. This challenges the industry norm of evaluating sponsored search effectiveness solely through CTR, highlighting that brand awareness effects can accumulate through exposure and may therefore be systematically underrepresented by behavioural metrics alone.

2.5 SEO, UX Design, and the Interface Context of Sponsored Search

Search engine optimisation (SEO) and user experience (UX) design have been treated as distinct subjects with often conflicting goals (Wang, 2023). While SEO focuses on keyword density, metadata structuring, and technical signals for algorithmic ranking, UX design values intuitive navigation, clear information structure, and user satisfaction. Combining these two domains is not just desirable, but necessary for creating interfaces that balance both the user experience and the business goals.

The challenges are directly applicable to sponsored search design (Wang, 2023). Tension becomes evident between SEO requirements and UX principles in sponsored result interfaces. The dynamic nature of both algorithmic signals and user behavioural expectations further complicates interface design. An advertisement format optimised for current ranking signals and user preferences may become suboptimal as these factors evolve. A practical foundation for the iterative improvement of sponsored search interfaces provided by the framework centres on user research, on-page SEO optimisation, UX design principles, responsive design, and continuous performance monitoring (Wang, 2023).

This framework, when applied to the context of SLs, supports the argument (Lu et al., 2017) that contextually rich advertisement content can achieve the central processing route and strengthen cognitive trust. However, from a UX perspective, this content must be presented in a manner that minimises cognitive load, employs clear structure and ensures the sponsored nature of results is communicated transparently without risking the user's search experience.

2.6 The Platform Ecosystem Perspective: Cross-Platform Advertising and Organic Ranking Authority

A dimension largely missing from the foundational sources, but increasingly relevant in understanding the contemporary e-commerce ecosystem, is the role of cross-platform advertising in shaping the information environment. This paper introduces this perspective as a fifth theoretical lens. Social media platforms amplify brand visibility through algorithmically curated content feeds, influence marketing, and visually engaging product presentations. When a product or a brand gains significant exposure through external platforms, the secondary effects are often reflected in users' search behaviour. Users subsequently search for these brands of products on SE, thereby increasing the branded search, which might lead to an improved organic ranking. This dynamic has been observed in practitioner SEO discourse; sometimes it is described as a feedback loop between social exposure and search performance.

The entire premise of organic search is to provide merit-based ranking, thus giving an opportunity to small businesses with relevant content to compete with big corporations with higher budgets to promote their products and increase accessibility. This dynamic favours brands with the resources to invest in cross-platform promotion, which can indirectly strengthen their organic ranking. As a result, organic visibility may increase, reflecting not only relevance and content quality but also the prior exposure shaped by the external platforms. The smaller brands with lower budgets, but highly relevant products, might never get visibility, leaving users with fewer and even less-relevant options.

This perspective also intersects with the trust framework proposed by Lu et al. (2017). Users assign a higher credibility to organic results because they are perceived as less commercially influenced. However, if organic rankings are partly shaped by promotional activity across platforms, this perception of neutrality may be incomplete. In such cases, the emotional trust users place in search engines as unbiased intermediaries may rest on an imperfect understanding of how visibility is constructed. Over time, increased awareness of these dynamics could challenge the perceived legitimacy of the search results page as a neutral information environment.

3 A Conceptual Framework: Sponsored Search as an Attention, Trust, Relevance, and Ecosystem System

3.1 Attention Allocation in Hybrid SERPs

Search engine result pages present the user with an integrated visual landscape whereby the paid and organic search results are distinguished primarily based on labelling and minimal differences in format. Users visiting these pages have to focus on the things that are important. They have to choose what to look at because they do not have a lot of time. The search results are different because some are chosen by a computer program and others are chosen because someone paid for them. Herbert Simon observed that when there is a lot of information, it is hard to know what to pay attention to. This means people have to be careful about what they choose to look at. (Danescu-Niculescu-Mizil et al., 2010).

There is a significant difference in terms of the degree to which users recognise and react to sponsored content labels. According to Lu et al.(2017), organic links are preferred statistically significantly in more than 82% of searches where content quality is equalised, whereas Danescu-Niculescu-Mizil et al. (2010) show that there is a consistent negative bias towards sponsored links when both links lead to the same website. Overall, it shows that there is some attentional asymmetry favouring organic links prior to trust formation in sponsored content.

Another important feature of the user behaviour identified by Danescu-Niculescu-Mizil et al. (2010) relates to the classification of the query into a navigational (when a known destination is searched for) or a non-navigational (either informational or transactional). Navigational queries are characterised by a high level of attentional competition for sponsored results, with one organic result usually meeting the search need of the user. As such, only incidentally placed advertisements providing complementary information will be noticed by the user. Non-navigational queries have a broader need, which is met incompletely with any organic result, and leave room for the advertisement to be seen.

3.2 The Pathway of Trust Meditation

Using the works by Lu et al. (2017), Aiello et al. (2016) and Danescu-Niculescu-Mizil et al. (2010), a model for trust mediation can be developed from two levels: the system levels, where the process of selecting advertisements and their ranking takes place, and the interface levels, which involve the way these advertisements are displayed to users.

Under the system level, relevance quality, which is defined as the match between the semantics of an advertisement content and the user's query intent, will act as the main trigger for rational trust formation under the central process model. If the sponsored search system ranks advertisements according to their contextually relevant nature, then people using central processing mode will feel that the system has enough competence and integrity. But as shown in the study by Aiello et al. (2016), ranking models that depend on eRPM may fail to achieve this task, since advertisements that have a very low click-through rate but are highly relevant get systematically de-prioritised.

Interface-wise, the reputation of the search engine affects the emotions of the user regarding the peripheral aspect. The concept of trust can be transferred from one entity to another; thus, the users' positive emotions regarding the search engine will be transferred to its ads. The feelings of security and confidence of the users with regard to the search engine might allow them to exhibit partial trust toward the ads shown by the search engine (Lu et al., 2017).

User attitudes play a role in shaping trust, especially when it comes to avoiding ads. Users with negative feelings toward ads may not trust them, no matter how good the platform's reputation is. They might even ignore ads that are relevant to them if they do not like ads in general. For these users, making ads more relevant or making the interface look more credible might not be enough to change their minds about ads. Users with a dislike for ads may resist trusting them and may not even think about ads in a careful way, even if the ads are very relevant to them.

3.3 The Platform Ecosystem as a Pre-SERP Trust Primer

The platform ecosystem approach offers a new dimension in the trust mediation process, which is the pre-SERP stage. The pre-SERP dimension takes place through exposure to the brand before the search experience through cross-platform interaction. In case a consumer is exposed to the brand outside of the search environment through various marketing activities such as promotions or influencer content, then the consumer forms impressions regarding the brand that guide their perception of SERPs.

This SERP priming process works through both trust channels highlighted by Lu et al. (2017). Via the peripheral channel, the process acts as a heuristic cue, thus promoting emotional trust as people become more comfortable interacting with products that they have seen before. The process works via the central channel as it provides information for making judgments about sponsored content.

Moreover, the relationship between cross-platform exposure and search activity also means that cross-platform advertisements have the potential to affect organic ranking authority as well. The increased branded search interest, brought about by prior exposure, affects the organic prominence of these entities, thus helping in determining what the SERP composition will be like. Hence, the placement of the brand in organic rankings is affected not only by relevance but also by exposure as well.

From the point of view of the users, this process seems to be transparent to them. Search results which come organically are normally seen as being objective and purely determined by algorithms, even if the visibility of those results depends to some extent on previous marketing campaigns conducted by the same company.

3.4 Decision-Making and Response Outcomes

The outcomes from user responses to sponsored searches involve both attitudinal consequences, like brand awareness, brand recall, and formation of attitudes, and behavioural consequences like click-through rate (CTR). Lu et al. (2017) have found that cognitive trust and emotional trust have an effect on the outcomes, with cognitive trust having a greater effect than emotional trust.

The implication from the research indicating that attitudinal effects can occur without the intervention of clicks is quite significant in terms of assessing the success of sponsored search. Being exposed to the advertisements alone can create recall and even affect the attitudes of users; this could imply that metrics measuring just clicks are insufficient.

The categorisation of outcomes into responsive and incidental, proposed by Danescu-Niculescu-Mizil et al. (2010), adds more depth to the understanding of outcomes in terms of outcome modelling. Whether the advertisement responds directly to the needs of the user or provides additional value to the user has a bearing not only on the probability of click-throughs but also on the kind of value being provided. An incidental ad appearing within a navigational query setting can create positive brand impressions without necessarily leading to clicks.

This platform ecosystem approach makes it even harder to attribute the success of the strategy even more. When an individual interacts with the sponsored listing from a brand that was encountered elsewhere through the platform, it is hard to identify the effect of each medium. Current attribution techniques, especially those that employ last click methodology, might unduly weigh the importance of sponsored listings.

4 Discussion

4.1 Benefits of Sponsored Search for Users and Markets

If the sponsored search works well, then there is a mutual benefit for various stakeholders. Users searching for information or transactions will have their interests satisfied by the contextually appropriate advertisements that enable them to identify relevant commercial alternatives. According to the responsive advertising model put forth by Danescu-Niculescu-Mizil et al. (2010), advertisements aligned with the intention of the user increase the informational richness of the page instead of conflicting destructively with non-sponsored content.

For advertisers, sponsored search offers a quantifiable means of reaching people who are searching for solutions, and this cannot be achieved by any other form of digital marketing. While display and social media advertising show their ads regardless of whether the user is looking for information on something, sponsored search shows its ads when the user is actively trying to find answers to his/her questions. For small businesses and emerging firms, this method gives them access to reach levels of popularity that would take much longer if the business relied on organic search.

4.2 Structural Tensions and Ethical Considerations

The key challenge in the realm of sponsored search involves the disparity between the optimisation goal of the eRPM ranking algorithm, which aims at maximising expected revenue, and the informational goal of the user, who seeks access to the most relevant and credible content. According to Aiello et al. (2016), this disparity can cause quality problems in the form of low visibility of relevant ads and the dominance of ads with high click-through rates even though they have lower relevance levels.

Non-commercial bias, according to Danescu-Niculescu-Mizil et al. (2010), wherein users show a preference for organic results even if there are no significant differences in terms of quality content, can be seen as a rational choice in dealing with uncertainties on the motivations underlying sponsored results. This is actually a cognitive shortcut for survival in a setting wherein the quality and neutrality of sponsored

results are not certain. But according to the platform ecosystem approach, the non-commercial heuristic will become more relevant in an organic result environment that is partly commercialised.

The labelling of sponsored results is an associated issue that raises an ethical question. Although the “Ad” or “Sponsored” label is common, it might not be enough to ensure that people can distinguish themselves when it comes to paid and organic results, especially when there is no time left for thinking and when dealing with inexperienced users. According to Lu et al. (2017), users who do not know how the sponsored link works will evaluate the organic and sponsored results based on the same approach, meaning that the labels might not always be considered.

4.3 Interface Design as a Mediating Variable

According to Wang (2023), one of the critical mediators between the logic behind ranking and the outcome of such actions is interface design. When it comes to sponsored search, it means that user experience depends not only on how advertisements are selected but also on the hierarchy of visuals, information load, labels' importance, and the connection between sponsored and organic ads.

Lu et al.'s (2017) discovery that contextual relevance is the most powerful determinant of cognitive trust ($\beta = 0.713$) has practical applications for interface design. In other words, sponsored search engines must encourage users to produce ads that are rich in contextual details via design policies and advertisement formats. The use of structured advertisement extensions such as seller ratings, price, availability, and service details can convert sponsored search links from mere navigation tools into decision aids, resulting in central route processing and cognitive trust.

Moreover, responsive and mobile-friendly designs are becoming increasingly critical with more users engaging with searches via mobile or conversational mediums. The effects on user attention seen by Danescu-Niculescu-Mizil et al. (2010) in desktop settings may be different in mobile settings because of a more restricted space to choose between natural and paid listings. Further investigation in future studies needs to investigate how the type of search query, device, and the nature of the medium impact user attention and trust.

5 Implications for Research, Practice, and Policy

5.1 Implications for Search Engine Operators

One suggestion made by search engines for consideration is the adoption of a clear relevance model within their paid search rankings as an addition to or partial replacement of eRPM-based optimisation processes. It has been proven by evidence found by Aiello et al. (2016) that the inclusion of relevance models in ranking processes leads to improved results both in terms of user experience and business-oriented gains, thus proving the existence of a possible gap in current perceptions regarding the compromise between user experience and business goals.

A second area to be considered involves transparency tools. Enhanced transparency and awareness in relation to the labelling of sponsored results, particularly in light of the findings of Lu et al. (2017), who note that these labelling tools are often overlooked by users, might lead to more responsible trust decisions on the part of the latter. The issue becomes more difficult when extending this concept beyond the SERP, owing to platform interactions.

5.2 Implications for Advertisers

According to the two-process theory of trust developed by Lu et al. (2017), specific steps can be undertaken by advertisers to create effective campaigns. Advertisements which make use of relevant information relating to a particular product (for instance, structured information about the product, ratings of sellers, etc.) can stimulate cognitive trust and thus achieve better results compared to generic keyword advertisements. The distinction between responsive and incidental advertisements made by Danescu-Niculescu-Mizil et al. (2010) implies that advertisements should be designed depending on the type of the search query: responsive advertisements are preferable in non-navigational searches.

In general, cross-platform visibility must be seen as an integral process rather than a series of individual investments in marketing. Actions aimed at increasing visibility on multiple platforms may drive branded search traffic, thus improving the visibility of content organically and adding credibility to sponsored content. In this case, the attribution model must evolve and go beyond individual platform evaluations to include cross-platform effects.

5.3 Implications for Policy

Policy considerations also require that the impact of the dependence on click history for determining rankings be assessed. Well-established advertisers can leverage their performance history to gain structural advantages over new competitors, irrespective of their relevance or the superior quality of their products. Such an approach would reduce the function of sponsored searches as a competition-enhancing channel. Possible policies might entail making ranking criteria more transparent or introducing measures through which new advertisers are judged by relevance before obtaining enough click information.

However, the platform ecosystem dynamics at a broader level also have significant implications for regulation. If organic search placement depends to some extent on past promotion experience on different platforms, organic rankings can no longer be considered completely separate from business operations. It thus undermines the usefulness of the distinction between organic and sponsored listings as a means of protecting consumers. In formulating regulatory policy, it becomes imperative that regulators look beyond search listings to the larger ecosystem of cross-platform influence.

6 Limitations and Directions for Future Research

This paper has a way of looking at things, and that is both its main strength and its main weakness. It looks at what other people have found out and tries to make sense of it instead of doing its own research. This means it is limited by what those other studies found out and the situations they were looking at. For example, Lu et al. Did an experiment in 2017 where they had students use search engine results pages in a lab. This might not tell us what would happen with people using search engines in life. The data that Danescu-Niculescu-Mizil et al. Looking at it in 2010 is from a time and place, namely Yahoo! Search, and the things they learned from it might not work today. The way they figured out what makes something relevant to a search might not be the same in different situations. Aiello et al. In 2016 came up with a way to figure out what makes something relevant. They only tested it in one specific way, which might not work for all kinds of searches or for people from different backgrounds. The paper about search engines, like the one by Lu et al., is limited by what search engines, like the one by Lu et al., can do. The search engines, like the one by Lu et al., have limitations, like the paper about search engines like the one by Lu et al.

The idea of how platforms work is still being figured out because we do not have enough information that shows how advertising, on multiple platforms, affects how things are ranked naturally. We need to do research to see if what we think is happening really is. We should watch some brands over time. See how people search for them, where they show up in search results and what they do to promote themselves on different platforms. It would be good to do some experiments to see how people react when they see a brand they have seen before. We can do this by showing people the search results, but changing whether they have seen the brand before and see if that makes them more likely to trust and click on it. This will help us understand how people decide what to click on when they are searching for something.

Further avenues for research include longitudinal studies on the evolution of users' trust in sponsored search in light of their increased familiarity with labeling conventions; the relationship between the navigational vs. non-navigational distinction and different devices and new search paradigms; the potential impact of integrating AI-generated content on search interfaces on the processes of attention, trust, and relevance; and comparisons across cultures of the development of cognitive and emotional trust under differing levels of exposure to online advertisements.

7 Conclusion

Sponsored search plays a role in digital information systems. It helps keep search services free by making money from ads, it acts as a channel for businesses to reach customers, and it also influences how users think, choose and decide on things. Just focusing on making money from sponsored search is not enough. We need to design systems that serve users over time. Sponsored search affects how users form opinions and make choices. So we must consider its impact on users, not revenue. Understanding sponsored search in a way helps create better systems. It is crucial for long-term success.

Through an integration of four key areas of research, the current paper has posited that the performance of sponsored search is determined through a variety of interdependent factors. Attention allocation depends on interface design, the form of queries made, and whether advertisements are responsive or incidental; trust development occurs through cognitive and affective routes, which differ in their susceptibility to contextual relevance, reputation of the search engine, and user predisposition toward ad avoidance; and relevance is a constant concern in search results, although click-based ranking strategies fail to solve it. These four dimensions may be supplemented with a fifth factor – namely, the ecosystem approach to the

platform, where the dynamics of the first four factors spill over into the realm of cross-platform attention priming and authority ranking.

In summary, the alignment of these factors indicates an ideal scenario for sponsored search where transparency, sensitivity to relevance, responsiveness to the variety of search intents, and proper contextualisation within the larger ecosystem of digital advertising are integral considerations. Achieving this ideal would involve collaborative work by search engine providers, advertisers, interface designers, policymakers, and other stakeholders, as well as ongoing empirical studies that keep abreast of developments in the dynamic environment of internet searching.

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